CREDIT RISK / MODEL RISK

Model Validation & Loss Sensitivity

A reproducible demonstration of PD development, out-of-time validation, statistical uncertainty, challenger benchmarking, temporal backtesting, and expected credit loss sensitivity.

0.763	0.739-0.789	0.378	0.098
ROC-AUC	**AUC 95% CI**	**KS**	**Brier**
2025 OOT	500 bootstraps	Discrimination	Probability error

1. Validation design

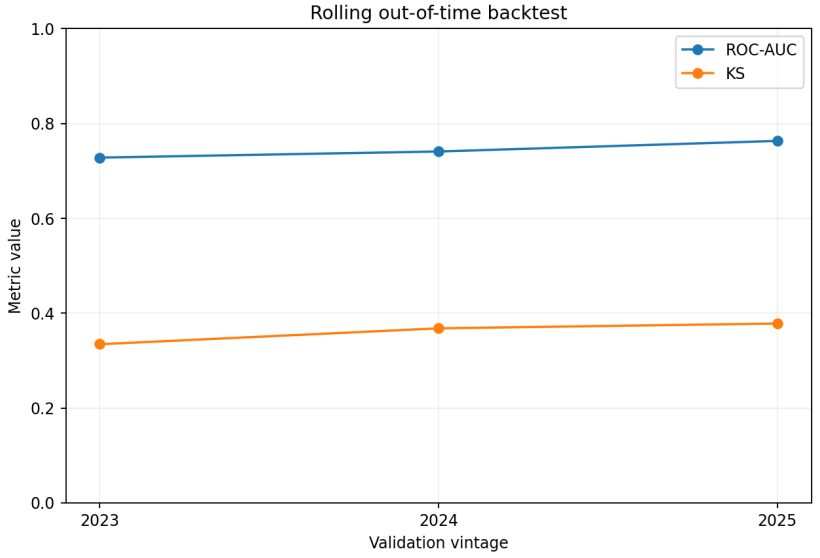
The model estimates 12-month probability of default using seven borrower-level variables. Development uses 8,932 observations from 2022-2024; the 2025 vintage (3,068 observations) is held out for final validation. Automated data controls pass 8/8 checks before model fitting.

2. Challenger and uncertainty

The logistic primary model achieves AUC 0.763, compared with 0.697 for the shallow decision-tree challenger. The paired-bootstrap AUC difference is +0.066, with a 95% interval of +0.045 to +0.086. This quantifies uncertainty in the model comparison rather than relying on two point estimates.

3. Rolling out-of-time backtest

Vintage	AUC	KS	Brier	Predicted PD	Observed DR
2023	0.728	0.335	0.086	11.06%	10.40%
2024	0.741	0.368	0.089	10.76%	10.93%
2025	0.763	0.378	0.098	10.85%	12.81%

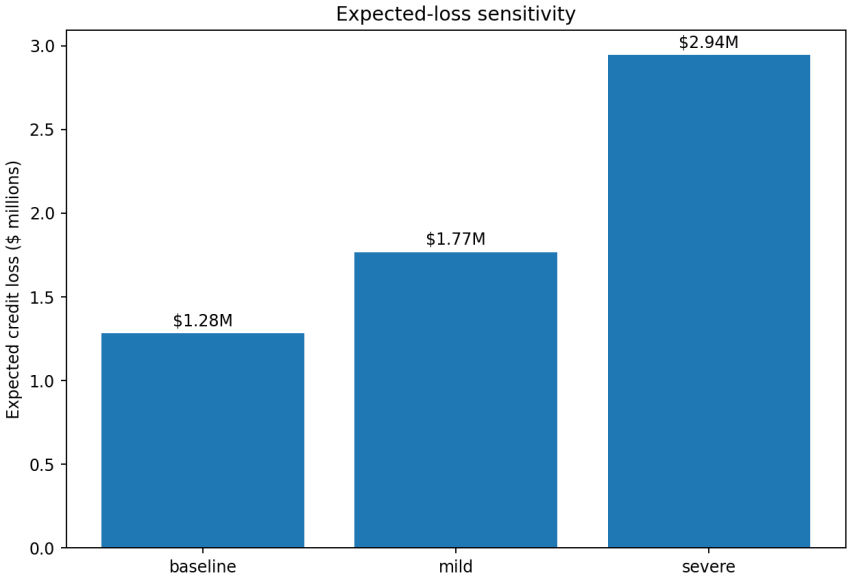


Monitoring signal: 2025 mean predicted PD is 10.85% versus a 12.81% observed default rate, a 1.95% absolute shortfall. Discrimination remains useful, but calibration has deteriorated.

Expected Credit Loss Sensitivity

The synthetic portfolio includes exposure at default (EAD) and loss given default (LGD) fields that are excluded from PD model inputs. They are used only after scoring to translate PD into financial loss using ECL = PD x EAD x LGD.

Scenario	Mean PD	Mean LGD	Expected loss	Change vs base
Baseline	10.85%	41.48%	$1.28M	-
Mild	13.40%	46.48%	$1.77M	+37.9%
Severe	20.29%	51.48%	$2.94M	+129.7%



4. Validation conclusion

The logistic model is retained as the preferred benchmark because it materially outperforms the challenger and maintains stable discrimination across three rolling validation vintages. It is not production-ready: 2025 calibration underpredicts realized default, and baseline expected loss ($1.28M) is below the synthetic realized loss ($1.51M). A production decision would require recalibration, additional history, independent validation and broader governance testing.

5. Limitations

- Synthetic data: no real borrowers or proprietary bank data are used.
- Illustrative EAD/LGD and stress assumptions: values are not calibrated to a real portfolio or supervisory path.
- Limited history: three backtest vintages demonstrate method, not production sufficiency.
- No causal or fair-lending conclusion: coefficients are predictive associations and the project is not a regulatory review.

Decision-use note: educational portfolio project only. Not a lending policy, regulatory validation, financial advice, or representation of any bank's production models.